

JING HUANG

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EDUCATION

Purdue University

Ph.D., Educational Psychology and Research Methodology (Advisor: Hua-Hua Chang)

Aug. 2023 – Present

West Lafayette, USA

Purdue University

M.S., Statistics (concurrent with the Ph.D.)

Jan. 2026 – Present

West Lafayette, USA

East China Normal University

M.Ed., Educational Psychology

Sep. 2020 – Jun. 2023

Shanghai, China

East China Normal University

B.S., Statistics

Sep. 2016 – Jun. 2020

Shanghai, China

RESEARCH INTERESTS

Continuous, real-time monitoring of live test administration, including computerized adaptive testing and longitudinal assessment:

- Sequential detection of aberrant examinee behavior (item preknowledge, low effort, and AI assistance)
- Real-time intervention for recovering ability estimates
- Pool maintenance through item parameter drift supervision, compromised item detection, and real-time online calibration

PUBLICATIONS

Summary: 10 peer-reviewed journal articles published (**5 as first author**); 5 manuscripts under review or revision (2 as first author, 1 as co-first author); 2 book chapters; 12 conference presentations (11 as first author).

* *Equal contribution.*

Peer-Reviewed Journal Articles

1. **Huang, J.**, & Chang, H.-H. (2026). Real-Time Monitoring of Item Parameter Drift in CD-CAT via Adaptive CUSUM Charts. *Psychometrika*, 1–61. [doi]
2. **Huang, J.**, & Chang, H.-H. (2026). A Cognitive Design System Approach to Enhancing Cognitive Alignment and Difficulty Control in Gen-AI Item Generation. *Applied Measurement in Education*, 1–21. [doi]
3. **Huang, J.**, Zhang, Y., Morphew, J. W., Nissen, J. M., Van Dusen, B., & Chang, H.-H. (2025). Two-phase Content-balancing CD-CAT Online Item Calibration. *Journal of Educational Measurement*, 62(4), 787–808. [doi]
4. **Huang, J.**, Xin, Y. P., & Chang, H.-H. (2025). The Application of Machine Learning to Educational Process Data Analysis: A Systematic Review. *Education Sciences*, 15(7), 888. [doi]
5. Mo, F., **Huang, J.**, Yang, Y., Özen, Z., Maeda, Y., & Olenchak, F. R. (2025). Undergraduate Students' Learning Outcomes with ChatGPT: A Meta-Analytic Study. *Computers and Education: Artificial Intelligence*, 100536. [doi]
6. Zheng, C., Liu, J., Li, Y., Xu, P., Zhang, B., Wei, R., Liu, B., & **Huang, J.** (2024). A 2PLM-RANK Multidimensional Forced-Choice Model and Its Fast Estimation Algorithm. *Behavior Research Methods*, 1–26. [doi]
7. Yang, X., **Huang, J.**, Chen, X., & Su, X. (2022). A Conceptual Analysis of Conceptual Understanding. *Exploring Education Development*, 20, 54–68. (in Chinese)
8. Li, Q., **Huang, J.**, & Xu, X. (2021). Relationship Between COVID-19 Infection Pneumonia and Traditional Chinese Medicinal Materials. *Chinese Journal of Medical Library and Information Science*, 30(01), 17–24. (in Chinese)

9. **Huang, J.**, & Xu, X. (2020). Research on the Influential Factors of Public Perception and Attitudes to COVID-19: Based on the Citizen Survey of Shanghai. *Library Journal*, 39(12), 98–106. (in Chinese)
10. Xu, X., & **Huang, J.** (2020). Research on Executives' Risk Perception and Behavior Choice to COVID-19. *Library Journal*, 39(12), 117–127. (in Chinese)

Manuscripts Under Review or Revision

1. Chang, H.-H., **Huang, J.**, Zhang, Y., Wang, X., Le, V., Mehrabi, A., Salazar, A., Morpew, J. W., Nissen, J. M., & Van Dusen, B. (revised and resubmitted). How We Know What Students Know: Rethinking College STEM Assessments with CD-CATs. *Psychometrika*.
2. **Huang, J.**, Zhang, J., & Chang, H.-H. (under review). A Dual-Dimensional LLM Framework for Automated Item Incidental Content Similarity Analysis in Large-Scale Assessments. *Applied Measurement in Education*.
3. **Huang, J.**, Guo, X., Richburg, A., Kim, S., Jiang, Y., Dallas, A., & Goodman, J. (under review). Leading Signals in Compromised-Item Monitoring: When Can Pre-Administration Vulnerability Improve Detection? *Educational and Psychological Measurement*.
4. Tao, Y.* , **Huang, J.*** , & Yang, X. (revised and resubmitted). What Drives Students' Everyday Creativity? *Thinking Skills and Creativity*.
5. Zhang, J., Liang, X., & **Huang, J.** (revised and resubmitted). A Multi-Agent Large Language Model Framework for Automated Q-Matrix Generation. *Psychometrika*.

Book Chapters

1. Zhang, Y., **Huang, J.**, & Chang, H.-H. (2025). Innovative Computerized Test Items. In *Encyclopedia of Social Measurement*. Elsevier. [doi]
2. Wang, S., **Huang, J.**, & Chang, H.-H. (2026). Computerized Adaptive Testing for STEM Learning. In *Artificial Intelligence for STEM Education Research: Advanced Methods and Applications* (pp. 87–111). Cham: Springer Nature Switzerland. [doi]

Works in Progress

1. **Huang, J.**, Fauss, M., Sinharay, S., & Chang, H.-H. Forensic Item Clones: Verifying External AI Assistance and Recovering Ability Estimates in Digital Assessment.
2. **Huang, J.**, & Cheng, M.-M. A Two-Stage Framework Integrating Psychometric Modeling and Process Data for Interpretable DIF Detection.

CONFERENCE PRESENTATIONS

1. **Huang, J.**, & Chang, H.-H. (2026, October). Forensic Item Clones: Verifying External AI Assistance and Recovering Ability Estimates in Digital Assessment. *NCME Artificial Intelligence in Measurement and Education Conference (AIME-Con)*, Pittsburgh, PA, USA.
2. **Huang, J.**, & Chang, H.-H. (2026, April). Adaptive CUSUM Charts for Real-Time Detection of Item Parameter Drift in CD-CAT. *Annual Meeting of the National Council on Measurement in Education (NCME)*, Los Angeles, CA, USA.
3. Wang, T., Yang, X., & **Huang, J.** (2026, April). A Mixture CDM for Mathematical Problem-Solving Strategies. *Annual Meeting of the National Council on Measurement in Education (NCME)*, Los Angeles, CA, USA.
4. **Huang, J.**, Zhang, Y., Wang, X., & Chang, H.-H. (2026, April). Optimizing Cognitive Diagnosis CAT: Addressing Practical Challenges through Psychometric Innovation. *Annual Meeting of the American Educational Research Association (AERA)*, Los Angeles, CA, USA.
5. **Huang, J.**, Zhang, J., & Chang, H.-H. (2026, April). LLM-Enhanced Automated Item Content Analysis Supporting CAT Applications. *Annual Meeting of the American Educational Research Association (AERA)*, Los Angeles, CA, USA.

6. **Huang, J.**, & Chang, H.-H. (2025, November). Generative AI-Enhanced Cognitive Design System Approach to Automatic Item Generation. *Purdue AI in P-12 Education Conference*, West Lafayette, IN, USA.
7. **Huang, J.**, & Chang, H.-H. (2025, July). Bridging Psychometrics and Generative AI: Automatic Item Generation for Verbal Analogical Reasoning. *International Meeting of the Psychometric Society (90th Annual Meeting)*, Minneapolis, MN, USA.
8. **Huang, J.**, Zhang, Y., Morphew, J. W., Nissen, J. M., Van Dusen, B., & Chang, H.-H. (2025, April). CD-CAT Online Item Calibration with Practical Constraints. *Annual Meeting of the National Council on Measurement in Education (NCME)*, Denver, CO, USA.
9. **Huang, J.**, Zhang, Y., & Chang, H.-H. (2024, November). Two-phase Content-balancing CD-CAT Online Item Calibration. *Ideas in Testing Research*, Chicago, IL, USA.
10. **Huang, J.**, Xin, Y. P., & Chang, H.-H. (2024, November). Machine Learning to Educational Process Data Analysis: A Systematic Review. *Purdue AI in P-12 Education Conference*, West Lafayette, IN, USA.
11. **Huang, J.**, Yang, T., & Yang, X. (2024, April). Taking a “Whole” Perspective: Exploring the Influence of Individual and Environmental Factors on Creative Performance. Roundtable, *Annual Meeting of the American Educational Research Association (AERA)*, Philadelphia, PA, USA.
12. **Huang, J.**, & Yang, X. (2022, April). Model-based Approaches to Classifying Students into Competence-based Proficiency Levels. Poster, *Annual Meeting of the American Educational Research Association (AERA)*, San Diego, CA, USA.

RESEARCH EXPERIENCE

Harold Gulliksen Psychometric Research Fellow — ETS

Aug. 2026 – Present

Mentors: Sandip Sinharay and Michael Fauss

- Develop sequential methods for real-time detection of aberrant examinee behavior using joint response and response-time data (dissertation research).
- Completed a manuscript on forensic item clones for verifying external AI assistance and recovering ability estimates.

Psychometric Research Intern — NCCPA

Jun. 2026 – Aug. 2026

National Commission on Certification of Physician Assistants

- Developed an LLM-based content-risk scoring rubric for certification items, rated by multiple large language models and evaluated with a generalizability study.
- Built a Bayesian change-point model that uses content-risk scores as prior information to detect compromised items in a longitudinal assessment program.
- First-authored manuscript under review at *Educational and Psychological Measurement*.

Graduate Research Assistant — Purdue University

Aug. 2024 – Jul. 2025

School of Engineering Education; NSF Grant No. 2141847

West Lafayette, USA

- Developed, maintained, and refined CD-CAT algorithms; led parameter estimation, item-selection optimization, online calibration, and CAT tuning procedures; conducted simulation studies.
- Supported mathematics cognitive-diagnosis development and psychometric refinement; performed psychometric analyses of operational [LASSO](#)-platform data.

Volunteer Consultant — Psychometric Investigation Laboratory

Dec. 2023 – Present

Purdue University

West Lafayette, USA

- Advise on statistical methods, research design, and data management.

Research Assistant — East China Normal University

Apr. 2021 – Jun. 2023

Suzhou Micro-Monitoring Project; “13th Five-Year Plan” Project

Shanghai, China

- Modeled creativity data from 4,833 students across 48 schools using Rasch analysis and multilevel models.
- Analyzed teacher–student interactions from classroom video data.

Research Assistant — Chinese Hospital Association & ECNU Clinical Pharmacist Training Evaluation	Jul. 2021 – Sep. 2021 Shanghai, China
Analyzed student-satisfaction factors using ordinal logistic regression.	
Research Assistant — Surveys and Data Center, ECNU Quality-of-Life Survey Studies	Sep. 2019 – Jun. 2020 Shanghai, China
Analyzed and cleaned large-scale survey data; co-authored two journal papers.	

TEACHING

Graduate Teaching Assistant — Purdue University College of Education	Aug. 2025 – May 2026 West Lafayette, USA
EDPS 32700 Classroom Assessment (Dr. Anne Traynor).	
Workshop Instructor — Psychometric Investigation Laboratory Purdue University	Dec. 2023 – Present West Lafayette, USA
<i>Using Machine Learning to Analyze Educational Data</i> workshop series.	

HONORS AND AWARDS

ETS Harold Gulliksen Psychometric Research Fellowship	Aug. 2026
NCME Graduate Student Works-in-Progress eBoard Competition, Award Winner	Apr. 2026
Psychometric Society Vector Travel Award	Jul. 2025
Ross Fellowship, Purdue University	Mar. 2023
Prior to Ph.D.	
National Scholarship (12 of 6461), Ministry of Finance & Ministry of Education, China	Oct. 2022
Excellent Graduate (101 of 6461), East China Normal University	Oct. 2022
Shu Qing Encouragement Scholarship (22 of 6461), East China Normal University	Oct. 2021
Third Prize, Contemporary Graduate Statistical Contest in Modeling (Team Leader)	Dec. 2020
Second Prize, COVID-19 Data Analysis Contest (Team Leader)	Jul. 2020
National Encouragement Scholarship (1 of 71), East China Normal University	Dec. 2019
Third Prize, Contemporary Undergraduate Statistical Contest in Modeling (Team Leader)	Nov. 2019
Shanghai Popular Science Education Innovation Award, Second Prize	Nov. 2019
Meritorious Winner, Interdisciplinary Contest in Modeling (ICM), COMAP	Feb. 2019
Third-Class Scholarship for Excellent Student, East China Normal University	2017, 2019

PROFESSIONAL SERVICE

Conference Reviewer	
NCME Annual Meeting — Individual Paper Proposals (10 proposals)	2027
AERA Annual Meeting — SIG-Computer and Internet Applications in Education (3 proposals)	2027
NCME AIME-Con — Innovation Demonstration & Paper/Poster Session Review Modules (7 proposals)	2026
Journal Reviewer	
<i>Psychometrika</i>	2025, 2026
<i>Journal of Computerized Adaptive Testing</i>	2025